

Ahmed Akram Amer

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate (GPA: 3.71/4.0) specializing in LLM applications, multi-agent systems, and GenAI solutions. Delivered 4+ production-ready AI systems using LangGraph, CrewAI, RAG pipelines, and Gemini 2.5 Flash — from enterprise Text-to-SQL platforms to intelligent procurement automation. Proficient in Python and the full AI/ML stack, with proven results including 95% model accuracy, 80% reduction in manual workflows, and sub-3s API response times.

PROJECTS

QueryGuard AI — Enterprise Text-to-SQL Platform | Python, Gemini 2.5 Flash, Streamlit, SQLite3, RBAC Apr 2025 – Jun 2025

- Built an enterprise-grade GenAI platform that translates plain English and Arabic questions into optimized, executable SQL queries with real-time results, enabling data democratization for non-technical business users.
- Implemented a Dual-Layer RBAC security system: Admins get full CRUD access while Standard Users are restricted to read-only via SQL Views — with DROP, DELETE, TRUNCATE, and ALTER blocked at both UI and LLM prompt levels.
- Designed a decoupled modular architecture (src/auth.py, src/ai_engine.py, src/db_manager.py) supporting one-line migration from SQLite3 to PostgreSQL, MySQL, or Amazon RDS — zero UI changes required.
- Powered by the 2026 Google GenAI SDK (Gemini 2.5 Flash), delivering accurate, sanitized SQL generation with schema-aware context injection and injection-resistant system prompting.

Multi-Agent Research & Writing System | Python, LangGraph, CrewAI, FastAPI, n8n, Groq LLaMA 3.3 Jan 2025 – Mar 2025

- Built a multi-agent pipeline that cut manual research time by ~80% and generated structured reports in under 2 minutes, connecting n8n, FastAPI, LangGraph, and CrewAI agents end-to-end.
- Designed a router node dynamically dispatching tasks to specialized agents: Research Analyst (SerperDev/Google Search) and Content Writer — with full AgentOps observability across all runs.
- Integrated Groq LLaMA 3.3 70B achieving GPT-4-comparable output without API costs; resolved 6+ integration challenges including Poetry cache conflicts and Arabic encoding in n8n.

Personal Agentic Assistant — Full-Stack RAG App | Python, FastAPI, LangChain, Pinecone, Google Gemini, Streamlit Nov 2024 – Jan 2025

- Built a full-stack agentic RAG application enabling users to upload PDF documents (50+ pages) and chat with them via a LangChain ReAct agent powered by Gemini 2.5 Flash.
- Backend: FastAPI and Pinecone vector database with LangSmith observability, achieving sub-3s response times. Frontend: Streamlit ChatGPT-like interface with real-time upload and multi-turn memory.
- Implemented CI/CD pipelines with GitHub Actions to automate testing and deployment of the FastAPI backend.

AI-Powered Procurement System — Multi-Agent | Python, CrewAI, LangChain, Groq LLaMA 3.3, Tavily, AgentOps Sep 2024 – Nov 2024

- Designed a 4-agent CrewAI pipeline automating the full procurement workflow: query generation, product discovery, web scraping, and Bootstrap HTML report generation — cutting manual research by 70%.
- Containerized with Docker for high-availability processing; agents compare prices across Amazon Egypt, Jumia, and Noon in real-time.
- Integrated AgentOps for monitoring token usage, cost, fail rate, and execution traces — achieving full observability across all agent runs.

Customer Support Chatbot — Multilingual RAG | Python, LangChain, Chroma, Groq, HuggingFace, Gradio Jul 2024 – Sep 2024

- Built a production-style RAG chatbot handling customer queries in Arabic and English from a PDF knowledge base using multilingual embeddings (intfloat/multilingual-e5-small) with MMR retrieval (k=6).
- Implemented hallucination control via strict system prompting — agent responds only from retrieved context or escalates to human support.

Deep Learning with PyTorch — Multi-Domain | Python, PyTorch, CNNs, Transfer Learning, Audio Classification Apr 2024 – Jul 2024

- End-to-end PyTorch pipelines for image classification (CIFAR-10, MNIST), audio classification, e-commerce clothing, and breast cancer detection — with regularization and inference-ready deployment.

ML Lifecycle Factory — ML Forge | Python, Flask, MLflow Oct 2023 – Feb 2024

- Architected a web platform automating the full ML lifecycle (ingestion to REST API deployment) as a factory pipeline, reducing overall ML cycle time by 40% via MLflow experiment tracking.

TECHNICAL SKILLS

AI Agents LangGraph, CrewAI, LangChain, LangSmith, AgentOps, RAG, ReAct, Prompt Engineering

LLMs & NLP Hugging Face Transformers, NLTK, SpaCy, Groq, Google Gemini, OpenAI

Deep Learning PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, Keras

Web & APIs FastAPI, Flask, Streamlit, Gradio

MLOps & Cloud MLflow, Docker, GitHub Actions (CI/CD), AWS (EC2, S3, SageMaker), Pinecone, Chroma, LangSmith

Data Pandas, NumPy, Matplotlib, Seaborn, Plotly, Dash

Languages Python, SQL (SQL Server, Oracle, SQLite)

Tools Git, GitHub, Jupyter, Google Colab, VS Code

EXPERIENCE

AI & Data Science Intern • DEPI | EYouth Jun 2024 – Dec 2024

- Engineered supervised ML models for predictive analytics, achieving 90% accuracy on structured datasets and reducing manual reporting time by 50%.
- Conducted EDA and feature engineering with Pandas and Seaborn across 10+ high-dimensional datasets, surfacing actionable business insights for stakeholders.
- Deployed supervised ML models as REST APIs using Flask and Docker, enabling real-time inference for business users and reducing inference latency in production environments.

EDUCATION

Bachelor of Electronic Engineering — Computer Science and Engineering

Menoufia University • GPA: 3.71 / 4.0 • Expected Graduation: October 2026

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI
- AI Engineer for Data Scientists Associate — DataCamp
- HCIA-AI — Huawei
- Python for AI and Data Science — IBM
- Data Scientist — DEPI & EYouth
- Applied Deep Learning — MaharaTech
- AI Camp — Sprints × Microsoft